

1.0 Introduction

The purpose of this report is to talk about my journey when I was studying Object Oriented Programming with Sir Nazmirul Hizzad. Throughout this subject I've done tutorials throughout week 1 till week 10. These tutorials helped me understand the concept of Object Oriented Programming even better, as the codes taught me how to solve real world situations better.

2.0 Knowledge Gained from each tutorial

1. Week 1: Introduction to Java

I learned the basics of Java, creating my first HelloWorld Java code

2. Week 2: Classes and Objects

I learned how to initiate objects as well as constructors.

3. Week 3-4: Inheritance

I learnt how to override as well as understanding what super keywords are.

4. Week 5: Encapsulation

I learned how to hide data and to add variables to the programme.

5. Week 6: Inheritance

I learn how to protect the access modifiers using superclass as well as learning the super() keyword.

6. Week 7: Abstraction

I learn how to add WashingMachine and Refrigerator as the subclass.

7. Week 8-9: Collection of File I/O

I learn how to implement the file writing (txt.file) and file reading

8. Week 10: GUI programming

I learned how to handle the event with ActionListener

3.0 Challenges Encountered

I was able to understand inheritance, encapsulation, abstraction, file I/O. Though GUI event handling was a bit challenging, same as time management and learning Git were also difficult.

4.0 How the challenges were overcome

I used diagrams, practiced using simple examples. I used online resources, set study hours, and created a Git cheat sheet.

5.0 Improvements in Java

I can design class hierarchies, apply OOP principles, write cleaner code, use Git, handle file I/O, create GUIs, and document professionally.

6.0 Understanding the OOP concept

I've gained an understanding of OOP concept.

Encapsulation protects data using private variables. Inheritance creates class hierarchies for code reuse. Polymorphism allows objects to take multiple forms. Abstraction hides implementation details.

7.0 Future Learning Plans

I will learn advanced Java, Spring Boot, build projects, prepare for OCP certification, and contribute to open source.

8.0 Conclusion

This course gave me a strong OOP foundation. I originally started with no OOP knowledge and I am able to now build complex Java programs.